COMP S492F/3920SEF
Machine Learning

Supervised learning III

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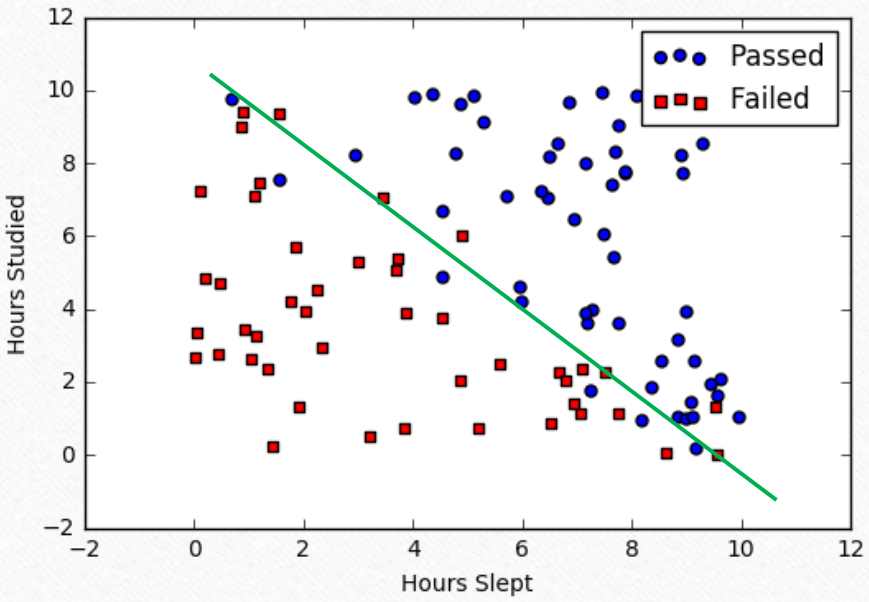
Previously on Logistic Regression:

How Logistic Regression Works

In logistic regression, a linear equation with optimized coefficients is to be found to give a decision boundary (in green) which can be used to classify the featured data into 2 classes as most accurate as possible (e.g., to predict a student to get a Pass or failure with 2 features: hours slept and hours studied).



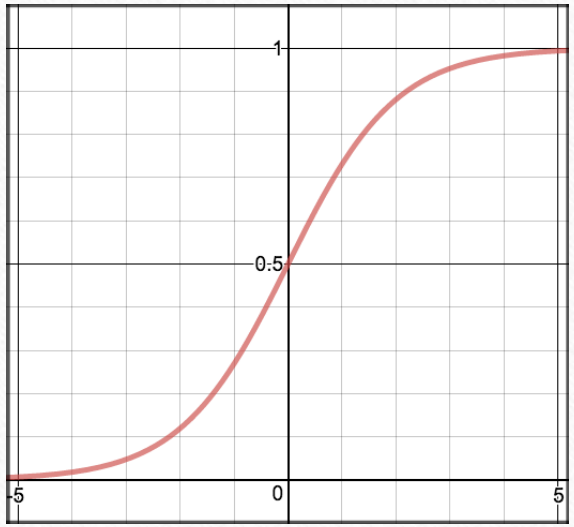
Steps taken (for binary classification):

1. Compute weighted outputs using a linear model equation (similar to linear regression).
2. Transform its output using the **Sigmoid function** to return a probability value which can then be used for mapping to the discrete classes.
3. Calculate the **Cross Entropy** of the predicted output and use it as the cost function.
4. Update the coefficients iteratively by using **gradient descent** as an optimization algorithm to minimize the cost function.
5. Stop until convergence or an acceptable error limit is reached.

Sigmoid Function and Threshold Value

- **Sigmoid function** (σ) is used to map predicted values to probabilities, i.e., map any real value into another value between 0 and 1:

$$h_\theta(x) = \frac{1}{1+e^{-\theta^T x}}$$



  - where $\theta^T x$ is the weighted output from the linear model

$$(\theta^T x = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n)$$

- Threshold value (= 0.5): with probability *p* as the sigmoid output:

$$p \geq 0.5, class = 1$$
$$p < 0.5, class = 0$$

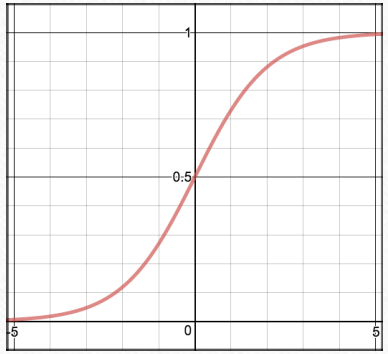
  - E.g. If the model returns 0.4 (<0.5), it believes there is only a 40% chance of being class 1, we would categorize this observation as class 0.

Cross Entropy and Cost Function

- Cross entropy is used instead of MSE as the cost function (J_{CE}) (log loss):

$$J_{CE} = -\underbrace{\frac{1}{m}\sum_{i=1}^{m}}_{\text{Sum over samples}} \underbrace{y^{(i)} \log\left(p\left(y^{(i)}\right)\right) + \left(1 - y^{(i)}\right) \log\left(1 - p\left(y^{(i)}\right)\right)}_{\text{Sum over categories/classes}}$$

In Practice, usually we use natural log

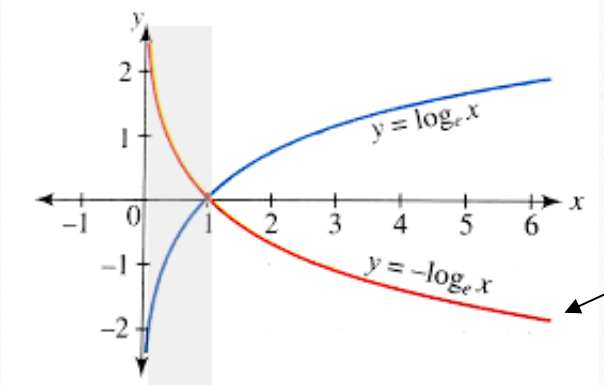


- In Logistic Regression $p(y_i) = h_\theta(x) = \frac{1}{1+e^{-\theta^T x}}$ based on the sigmoid function

$$J_{CE}(h_\theta(x), y^{(i)}) = \begin{cases} -\log(h_\theta(x)) & \text{if } y^{(i)} = 1 \\ -\log(1 - h_\theta(x)) & \text{if } y^{(i)} = 0 \end{cases}$$

where h_θ is the predicted probability, t is target (true label).

- For $y^{(i)}$ =1 : if h_θ=0 then J_{CE}=+∞ and if h_θ=1 then J_{CE}=0.
- For $y^{(i)}$ =0 : if h_θ=0 then J_{CE}=0 and if h_θ=1 then J_{CE}=+∞ .



Look at this curve

Given the following Logistic Regression problem, write the loss/cost function of it

Model:	$\theta = \{\theta_2, \theta_1, \theta_0\}$
Features (i^{th}):	$x^{(i)} = \{x_2^{(i)}, x_1^{(i)}, x_0^{(i)}\}$
Target (i^{th}):	$y^{(i)}$
# of training data:	m
Activatation function:	Sigmod fucntion
Cost function:	Cross entropy

Answer:

$$J_{CE} = -\frac{1}{m}\sum_{i=1}^{m} y^{(i)} \log\left(p\left(y^{(i)}\right)\right) + (1 - y^{(i)}) \log\left(1 - p\left(y^{(i)}\right)\right)$$

$$= -\frac{1}{m}\sum_{i=1}^{m} y^{(i)} \log(h_\theta(x)) + (1 - y^{(i)}) \log(1 - h_\theta(x))$$

$$= -\frac{1}{m}\sum_{i=1}^{m} y^{(i)} \log\left(\frac{1}{1 + e^{-\theta^T x^{(i)}}}\right) + (1 - y^{(i)}) \log\left(1 - \frac{1}{1 + e^{-\theta^T x^{(i)}}}\right)$$

$$where:\ \theta^T x^{(i)} = \theta_0 x_0^{(i)} + \theta_1 x_1^{(i)} + \theta_2 x_2^{(i)}$$

Suppose the order of all features in the model are 1

Gradient Descent and Update Rule

- The partial derivative of the sigmoid function is given by:

$$\frac{\delta}{\delta\theta_j} J_{CE}(\theta) = \frac{1}{m}\sum_{i=1}^{m}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)x_j^{(i)}$$

  - Note that this is the same as the MSE partial derivative, the only difference is the hypothesis function. A hypothesis function is the one that best describes the target.

- Update rule (same as that for linear regression):

$$\theta_j := \theta_j - \alpha\frac{1}{m}\sum_{i=1}^{m}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)x_j^{(i)}$$

Model Evaluation for Classification: Confusion Matrix

- A confusion matrix is used to show the number of correct/incorrect predictions made by the classification model compared to the actual outcomes (target value) in the data.
- Imagine there are 10,000 cases (with 100 positive cases only) for predictions by a binary classifier and the results are tabulated in a *confusion matrix*:

	Predicted Negative	Predicted Positive
Negative Cases	TN: 9,760	FP: 140
Positive Cases	FN: 40	TP: 60

There are 4 ways of being right or wrong:

- TN / True Negative: the case is negative and predicted negative
- TP / True Positive: the case is positive and predicted positive
- FN / False Negative: the case is positive but predicted negative
- FP / False Positive: the case is negative but predicted positive

- 3 performance metrics can be used for evaluation of the model performance:
  - **Accuracy** = (TN + TP) / (TN + TP + FN + FP) = (9760 + 60) / 10000 = 98.2%
  - **Precision** = TP / (FP + TP) = 60 / (140 + 60) = 30%
  - **Recall (Sensitivity)** = TP / (FN + TP) = 60 / 100 = 60%

Accuracy? Precision? Recall?

Accuracy is a great measure but only when you have a balanced dataset in which the numbers of positive and negative classes are almost the same.

Cancer prediction: Out of all the people that do actually have cancer, how many do we have identified?

- If we don’t want to fail to spot many people that actually have cancer, i.e. we are more concerned with a minimal false negative.
- Higher *recall* (quantity-concerned) is more important.

（宁枉勿纵 Quantity over Quality）

Document retrieval: Out of all the relevant documents, how many do we have retrieved?

- If we don’t want to retrieve many documents that are irrelevant, i.e. we are more concerned with a minimal false positive.
- Higher *precision* (quality-concerned) is more important.

（宁缺毋滥 Quality over Quantity)

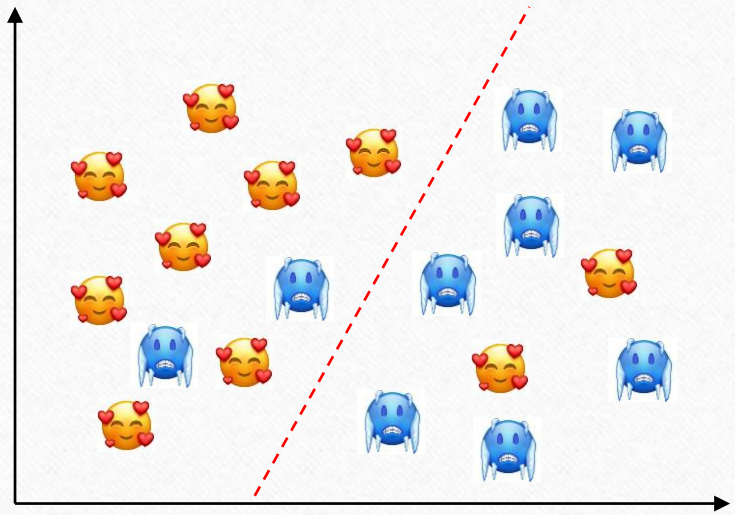
Precision and Recall Tradeoff

Definition: The problem that a binary classifier can only be adjusted to increase *precision* at the expense of *recall*, or vice versa.

TN: True Negative
TP: True Positive
FN: False Negative
FP: False Positive

$$\text{Precision} = \frac{\text{TP}}{\text{FP} + \text{TP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{FN} + \text{TP}}$$



- Ideally, if there is a perfectly separable data, both precision and recall can get maximum value of 1.0.
- In practical situations, noise exists in the dataset and the dataset is not perfectly separable.
- Some points of positive class might be closer to the negative class and vice versa.
- In such cases, shifting the decision boundary can either increase the precision or recall but not both (i.e., the increase of precision leads to the decrease of recall or the increase of recall leads to the decrease of precision).
- In other words, binary classifier may always misclassify some points.

Regularization in Logistic Regression

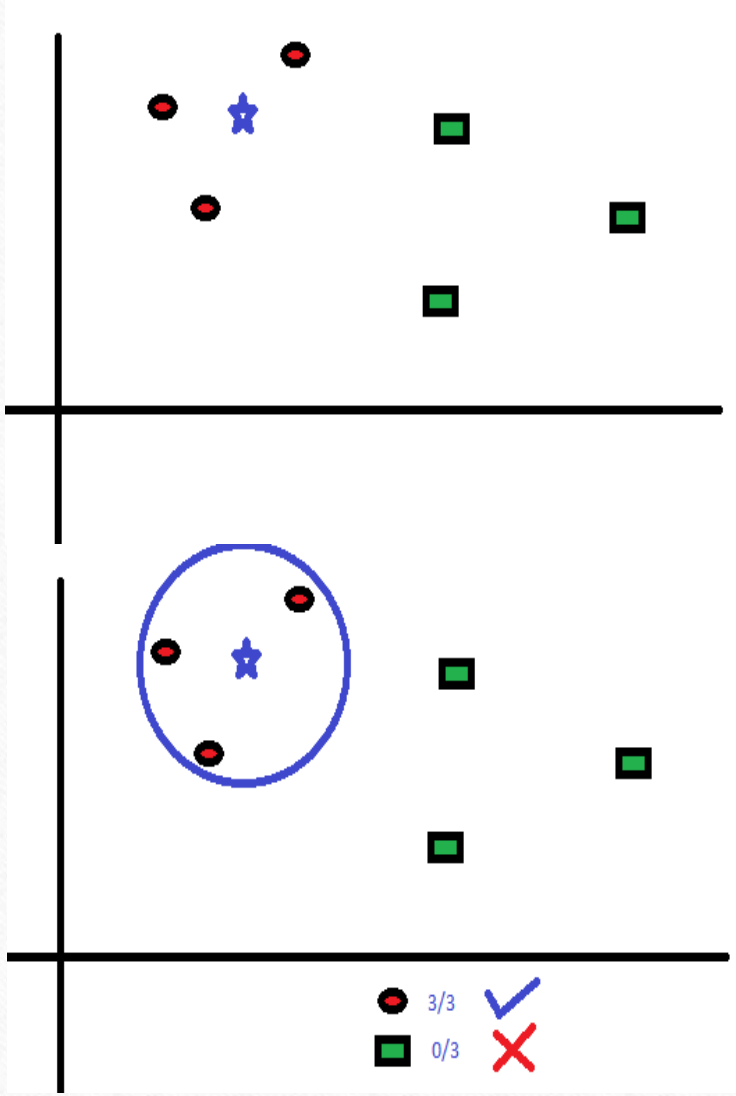
- Similar to linear regression, a function is added to penalize large values of the parameters. Most often the function is $\lambda \sum_{j=1}^{n} \theta_j^2$, which is some constant λ (the *regularization strength*) times the sum of the squared parameter values θ_j^2.

- The value of λ controls the trade-off between allowing the model to increase its complexity as much as it wants with trying to keep it simple. For example, if λ is very low or 0, the model will have enough power to increase its complexity (overfitting occurs) by assigning big values to the weights. If, on the other hand, we increase the value of λ, the model will tend to underfit, as the model will become too simple.

Previously on KNN:

How K-Nearest Neighbors Works (Example)

- The figure shows a spread of red circles (RC) and green squares (GS). You intend to find out the unknown class of the blue star (BS), which can be either RC or GS.
- The “K” in KNN algorithm is the number of the nearest neighbors we wish to take vote from. Let’s say K = 3.
- Hence, we will now make a circle with BS as center just as big as to enclose only three data points on the plane. The metric often used for distance measurement is the Euclidean distance, i.e., $dist(a,b) = \sqrt{(a_1 - b_1)^2 + \cdots + (a_n - b_n)^2}$.
- The three closest points to BS are all RC. By voting with a result 3/3 votes (i.e. 100%) for RC, we can say that the BS should belong to the class RC.



Worked Example of KNN

- We have data from a survey on a special paper tissue and objective testing with its two attributes (acid durability and strength) to classify whether it is good or not. Below are 4 samples:

	X1 = Acid durability (seconds)	X2 = Strength (kg/square meter)	Y = Classification
	7	7	Bad
	7	4	Bad
	3	4	Good
	1	4	Good
New paper tissue	3	7	?

- Now the factory produces a new paper tissue that passes the laboratory test with X1 = 3 and X2 = 7. Without another survey at a high cost, take a guess of the classification of this new paper tissue.

Worked Example (cont'd)

- Suppose K = 3.
- Now calculate the Euclidean distances between the query instance and all the sample instances. We can save time just by calculating the square distances instead.

X1 = Acid durability (seconds)	X2 = Strength (kg/square metre)	Square distance to query instance (3,7)
7	7	$(7 – 3)^2 + (7 – 7)^2 = 16$
7	4	$(7 – 3)^2 + (4 – 7)^2 = 25$
3	4	$(3 – 3)^2 + (4 – 7)^2 = 9$
1	4	$(1 – 3)^2 + (4 – 7)^2 = 13$

Worked Example (cont'd)

- Then sort the distance and determine the 3 nearest neighbors based on the K-th minimum distance to check their classifications.

X1 = Acid durability (seconds)	X2 = Strength (kg/square metre)	Square distance to query-instance (3,7)	Rank by minimum Distance	Included in 3 nearest neighbors?	Y = Classification
7	7	$(7 – 3)^2 + (7 – 7)^2 = 16$	3	Yes	Bad
7	4	$(7 – 3)^2 + (4 – 7)^2 = 25$	4	No	~~Bad~~
3	4	$(3 – 3)^2 + (4 – 7)^2 = 9$	1	Yes	Good
1	4	$(1 – 3)^2 + (4 – 7)^2 = 13$	2	Yes	Good

- Finally use simple majority of the classification of nearest neighbors to find the predicted class of the query instance. We have 2 good's and 1 bad. Therefore we can predict the new paper tissue to be in the 'good' class.
- KNN algorithm can also be used for regression other than classification problems. The only difference will be using averages of nearest neighbors rather than voting from nearest neighbors.

KNN Test Example

We have data from a survey on a car and objective testing with its three attributes (max_speed, power_efficiency and cost) to classify whether it is good or not.

Below are 5 samples:

	Max speed	Power_efficiency	Cost	Good/Bad
A	0.9	0.8	0.1	Good
B	0.5	0.1	0.8	Bad
C	0.7	0.2	0.7	Bad
D	0.9	0.1	0.6	Bad
E	0.8	0.7	0.2	Good

Now we have a new car with Max speed =0.6, Power_efficiency=0.9 and cost=0.3. Determine its class by KNN if (a) k=1 (b) k=3 (c) k=5

Answer will be upload in OLE week 5

The end